

Answer any 3 (1 and 4 are compulsory)

1. Dipu is trying to set up his new startup company's network, which consists of twenty devices, including a router connected to an ISP. He thought of cutting the cost of installing a DHCP server by sharing the DHCP server of his friend Zahin. Zahin works for another company with a DHCP server set up in the same building two floors down. However, to Dipu's surprise, his devices cannot receive any IPv4 configuration from the DHCP server. Identify the problem here and explain how it can be solved.

2. You can access and use the coffee shop's Internet when you sit at a coffee shop. No one gave you the coffee shop's network's IP address to have access. Explain how it was possible.

3. A network engineer uses the "ipconfig" command to visualize the current IP network configuration. From the displayed output, he notes down the following information:

IPv4 Address : 17.50.96.81

Subnet Mask : 255.255.128.0

He also identifies that the default-gateway of the network is the second usable host IP address of the network.

From the above scenario, determine the following (show necessary calculations):

I. The broadcast address

II. IP address of the default-gateway

4. Dipu was trying to communicate to a web server, where the connection is set up as shown in the figure below. The network the web server belongs to is a private network and R2 is configured to use NAT. **Interpret** how it will be possible for Dipu to communicate with the web server when it does not have a fixed public IP address assigned to it.

